# NutriAI

Personalised 7-Day Diet Planning powered by ML

Final Project | Option A | June 05, 2026

## EXECUTIVE SUMMARY

NutriAI is a production-grade automated diet planning application that generates a personalised 7-day meal plan in 2-3 seconds, strictly tailored to a user's clinical conditions, dietary preferences, allergen restrictions, and daily nutritional targets. The system integrates all five BAX-423 ML techniques (Sketching, Embeddings, Recommendation, Ranking, Streaming) across a 13,620-food USDA database. All four required test personas pass all six core capability checks. The app is deployed live on Streamlit Community Cloud.

## DATA PIPELINE & STATISTICS

### Pipeline Stages

| | |
|---|---|
| **1. USDA Ingest** | build_database.py - Foundation + SR Legacy + FNDDS |
| **2. Enrichment** | enrich_database.py - clinical tags, allergen flags, GI estimates |
| **3. Clinical Filter** | SQL WHERE: FODMAP, GERD, GI<=55, sodium cap, 50+ name guards |
| **4. ANN Retrieval** | 15-dim nutrient embeddings, cosine k-NN (k=120) via sklearn |
| **5. Bloom Filter** | Belt-and-suspenders allergen check, 0% false-negative rate |
| **6. Multi-obj Rank** | 5-dimension weighted scorer (calorie, gap-fill, diversity, priority, clinical) |
| **7. Streaming Output** | Python generator yields each meal live; Streamlit updates in real time |
| **8. Gap Feedback** | Day totals feed back into gap vector; next slot adapts to what |

### Key Statistics

| | |
|---|---|
| **Foods in DB:** | 13,620 |
| **Nutrient columns:** | 21 per food |
| **Data types:** | Foundation, SR Legacy, FNDDS |
| **Allergen filters:** | 11 types, Bloom-backed |
| **Clinical conditions:** | IBS, GERD, T2DM, Hypertension |
| **Diet modes:** | 4 (vegan, vegetarian, pescat., omni) |
| **BAX-423 techniques:** | 5 of 5 applicable |
| **Generation time:** | 2-3 s end-to-end |
| **Diversity index:** | 0.73-0.84 (Simpson's D) |
| **Personas passing:** | 4 / 4 (all 6 checks) |

## SIGNATURE DELIVERABLES

| | |
|---|---|
| **"Why Excluded":** | Every filtered food returns a clinical/allergen reason string (explainer.py) |
| **PDF + CSV Export:** | 10-page PDF plan | 3 CSVs (meals, daily summary, exclusions) + ZIP |
| **Sub-60s Generation:** | Typical cold start ~2-3s; warm (cached Bloom+embeddings) <1s |
| **Pass/Fail Table:** | All 4 personas x 6 capabilities - see Page 3 |

## BAX-423 TECHNIQUES & BENCHMARKS

### T1 - SKETCHING (Bloom Filter)

*Probabilistic Data Structures for allergen safety*

11 AllergenFilterBank instances (one per allergen type). Each filter: m = 1.2M bits, k = 7 hash functions (MD5 + SHA-256 two-hash trick). Built once at session start from the full 13,620-food database. Applied AFTER SQL allergen clauses - belt-and-suspenders: SQL handles the bulk exclusion; Bloom guarantees no allergen escapes through edge-case composite foods.

| Metric | |
|---|---|
| Build time (13,620 foods, 11 filters) | **873 ms (one-time per session)** |
| Lookup time per food | **97.8 mcgs (O(1), k=7 hashes)** |
| False-negative rate | **0.000% (by construction)** |
| False-positive rate | **~0.08% (harmless: only over-excludes)** |
| vs SQL-only safety | **SQL alone: ~0.1% allergen escape on complex composites** |

### T2 - EMBEDDINGS (Nutrient Vectors)

*Vector Representations for nutritional similarity*

15-dimensional nutrient embedding per food: [calories, protein, carbs, fat, fiber, calcium, iron, potassium, magnesium, zinc, B12, vit_D, vit_C, omega3_ALA, omega3_EPA]. Transform pipeline: weighted (B12=2.0, vit_D=1.8, iron=1.5 to amplify scarce nutrients) -> log1p (compress concentration outliers) -> L2-normalise (enable cosine similarity). Query vector = daily nutritional gap vector (fractional remaining RDA per nutrient, sodium excluded). Index: sklearn NearestNeighbors, cosine metric, brute-force, n_jobs=-1.

| Metric | |
|---|---|
| Index build (5,000 foods) | **~60 ms** |
| Query time (k=120) | **<5 ms** |
| Gap-fill score: ANN top-5 | **5.26 (nutrient relevance index)** |
| Gap-fill score: random 5 foods | **1.60 (baseline)** |
| ANN improvement over random | **+230% - retrieves foods 3.3x more aligned to gap** |

### T3 - RECOMMENDATION (Content-Based ANN)

*Recommendation Systems - adaptive gap-filling content filter*

Content-based filtering: the query IS the user's current nutritional state. After each meal is assigned, day_totals updates, compute_gap_vector() recalculates the fractional remaining RDA, and the next ANN query reflects what was actually consumed - creating a closed adaptive feedback loop across all 21 meal slots. k=120 candidates retrieved, then Bloom-filtered, slot-suitability-filtered, ranked.

| Metric | |
|---|---|
| Adaptive iterations per plan | **21 (gap vector recalculated after every meal)** |
| Candidates retrieved per slot | **120 (cosine ANN from candidate pool)** |
| Pool sizes (typical) | **1,782 (Mei) to 3,805 (Ravi)** |
| Coverage vs fixed-gap baseline | **Adaptive: 4/7 RDA days (Priya) vs 0/7 fixed** |

### T4 - RANKING (Multi-Objective Weighted Score)

*Ranking & Learning-to-Rank - five-dimension weighted scorer*

5 scoring dimensions combined as weighted sum -> final score [0,1]. W1=0.20 Calorie Fit: Gaussian on delivered_kcal/slot_target, sigma=0.5, with per-day budget guard (<=110% of daily target) and per-slot calorie density cap. W2=0.32 Nutrient Gap-Fill: fractional RDA fill weighted by relative gap urgency. W3=0.23 Diversity: Simpson's D penalty 1/(1+count*decay) per food group; legumes decay=0.30, vegetables=0.45 to prioritise nutritionally important groups. W4=0.17 Priority Micro: nutrient-calibrated thresholds per priority nutrient. W5=0.08 Clinical Bonus: condition-specific food group and nutrient rewards.

| Metric | |
|---|---|
| Diversity - multi-objective ranker | **0.816 (Simpson's D, Priya; range 0.78-0.87)** |
| Diversity - random selection baseline | **0.617 (same food pool, no ranking)** |
| Diversity improvement | **+0.199 (+32% better food-group spread)** |
| Calorie adherence (all personas) | **0/7 days exceed 110% of target (after budget guard)** |
| RDA coverage: Priya "days met" | **4/7 days (vs 0/7 before gap-fill reweighting)** |

### T5 - STREAMING (Progressive Plan Generation)

*Streaming & Real-Time Processing - live meal-by-meal delivery*

generate_plan_stream() is a Python generator. Each of 21 meal slots yields a typed event dict {type:"meal", day, slot, meal, elapsed_s} immediately after selection. Streamlit iterates the generator and updates the progress bar + live description for each event. The user sees meals appearing one-by-one rather than waiting for the entire plan to complete.

| Metric | |
|---|---|
| Time to first meal (cold start) | **~2.1 s (includes Bloom build + embedding index)** |
| Total plan time (cold start) | **~3.0 s** |
| Time to first meal (warm, cached) | **<200 ms (Streamlit @st.cache_resource)** |
| Events yielded per plan | **22 (21 meals + 1 "complete" event)** |
| vs batch (wait for all 21) | **User sees plan building live; 17% faster first result** |

## PERSONA PASS / FAIL TABLE - All 4 Test Personas x 6 Core Capabilities

### Priya (28y F, 1800 kcal)

| | |
|---|---|
| **Diet:** | Vegetarian |
| **Condition:** | IBS |
| **Allergen:** | Dairy |
| **Priority Micros:** | Iron, Calcium, Vit D |
| **Weight / Protein floor:** | 58 kg => >41 g/day protein |

### Ravi (38y M, 2200 kcal)

| | |
|---|---|
| **Diet:** | Non-veg / No pork |
| **Condition:** | GERD |
| **Allergen:** | Gluten (Celiac) |
| **Priority Micros:** | B12, Zinc, Mg |
| **Weight / Protein floor:** | 80 kg => >56 g/day protein |

### Mei (45y F, 1600 kcal)

| | |
|---|---|
| **Diet:** | Vegan |
| **Condition:** | Type 2 Diabetes |
| **Allergen:** | Tree Nuts |
| **Priority Micros:** | B12, Iron, Zinc |
| **Weight / Protein floor:** | 62 kg => >43 g/day protein |

### James (52y M, 2000 kcal)

| | |
|---|---|
| **Diet:** | Pescatarian |
| **Condition:** | Hypertension |
| **Allergen:** | Soy |
| **Priority Micros:** | Potassium, Mg, Omega-3 |
| **Weight / Protein floor:** | 88 kg => >62 g/day protein |

| Capability | Priya | Ravi | Mei | James |
|---|---|---|---|---|
| **C1 Clinical Condition Filtering** | **PASS** | **PASS** | **PASS** | **PASS** |
| *FODMAP, low-acid, GI<=55, DASH sodium cap* | | | | |
| **C2 Allergy Detection & Exclusion** | **PASS** | **PASS** | **PASS** | **PASS** |
| *SQL + Bloom filter; zero false negatives* | | | | |
| **C3 Dietary Preference Handling** | **PASS** | **PASS** | **PASS** | **PASS** |
| *Vegetarian / vegan / pescatarian / non-veg* | | | | |
| **C4 Diversity Engine** | **0.834** | **0.816** | **0.730** | **0.839** |
| *No food repeated; Simpson's D >= 0.70* | | | | |
| **C5 Macro & Micro Analysis** | **PASS** | **PASS** | **PASS** | **PASS** |
| *21 nutrients tracked; RDA gaps flagged at 80%* | | | | |
| **C6 Sub-60s Generation** | **3.0s** | **2.9s** | **2.1s** | **2.1s** |
| *Cold ~2s; warm (cached) <1s* | | | | |

## KEY OUTCOME METRICS (per persona)

| Metric | Priya | Ravi | Mei | James |
|---|---|---|---|---|
| Generation time | 3.0 s | 2.9 s | 2.1 s | 2.1 s |
| Candidate pool size | 1,960 | 3,805 | 1,782 | 2,480 |
| Diversity (Simpson's D) | 0.834 | 0.816 | 0.730 | 0.839 |
| Days >= 80% RDA (+) | 4 / 7 | 0 / 7 * | 1 / 7 * | 0 / 7 * |
| Calorie adherence | Avg 95% | Avg 97% | Avg 91% | Avg 94% |
| Days >110% calories | 0 / 7 | 0 / 7 | 0 / 7 | 0 / 7 |
| Iron >= 80% RDA daily | PASS 6/7 | - | - | - |
| Fiber >= 25g/day | - | - | 3-5 / 7 | - |
| Potassium >= 80% RDA | - | - | - | 4 / 7 |
| Sodium <= 1500mg/day | - | - | - | 0 / 7 ** |

*(+) "Days >= 80% RDA" excludes Vitamin D (primarily sunlight-dependent; food alone <5% of 600 IU RDA), Omega-3 ALA (plant sources excluded as non-standalone-meal items), and Sodium (a cap, not a floor). * Structural constraints: Ravi fiber 38g/day near-impossible with GERD+gluten-free; Mei calories/fat on vegan+T2DM+1600 kcal; James fat/fiber on pescatarian+hypertension+soy-free. ** James sodium: cumulative multi-slot overshoot; individual foods all pass is_high_sodium=0 filter.*

## ADAPTIVE LEARNING - CLOSED-LOOP GAP FEEDBACK

NutriAI implements within-plan adaptive learning: after each meal is assigned, compute_day_totals() aggregates actual nutrients delivered, and compute_gap_vector() recalculates fractional remaining RDA needs (Eq: gap[n] = max(0, (RDA - actual) / RDA)). The ANN query and the ranker's gap-fill score both receive this updated vector before selecting the next slot. This creates a 21-iteration closed feedback loop: each meal selection observes what was already consumed and adapts its priorities accordingly.

Impact: Priya's "days meeting 80% RDA" improved from 0/7 (fixed gap) to 4/7 (adaptive) after correcting the gap normalisation. James potassium improved from 1/7 to 4/7. Mei fiber improved from 1/7 to 3-5/7. The adaptive loop accounts for the largest single improvement in RDA coverage across all personas.

## DEPLOYMENT & SUBMISSION

| | |
|---|---|
| **Live App URL:** | https://nutriai-bax423-msba.streamlit.app (Streamlit Community Cloud) |
| **Tech Stack:** | Python 3.13, Streamlit, SQLite, scikit-learn, fpdf2, plotly, bitarray |
| **Database:** | 13,620 foods \| nutriai_foods.db \| ~10 MB SQLite \| in repo |
| **Setup:** | pip install -r requirements.txt -> streamlit run app.py |
| **Rebuild DB:** | python build_database.py (13 min, USDA API) + python enrich_database.py (24s) |
| **Tests:** | python test_personas.py - all 4 personas, 6 capability checks, ~5 s |
| **ZIP Contents:** | code/ (full src), data/ (nutriai_foods.db snapshot), brief.pdf, prompts.md, README.md |

## KNOWN LIMITATIONS (Honest Assessment)

**Ravi 0/7 RDA days:**

Fiber RDA of 38 g/day (male) is clinically near-impossible with GERD + gluten-free combined. The pipeline correctly avoids bran/psyllium (excluded as supplements) and acidic high-fiber foods (GERD triggers). This reflects real dietary constraint, not a pipeline failure.

**James cumulative sodium:**

The ranker excludes individually high-sodium foods (>400 mg/100 g) via is_high_sodium=0 SQL flag, but three moderate-sodium foods combined can exceed the 1,500 mg/day DASH cap. Fix: pass rolling day sodium into the gap vector as a soft penalty once budget > 60%.

**Vitamin D & Omega-3 ALA:**

Food-only Vitamin D achieves <5% of the 600 IU RDA for dairy-free profiles. Omega-3 ALA requires nuts/seeds/oils which are excluded as non-standalone meal items. Both are excluded from the "days met" UI metric with a footnote; supplementation is recommended.

**Calorie variation +/-10%:**

Daily calorie delivery varies +/-5-10% around the user target. A per-slot calorie density cap (scaled to user calorie target) and a per-day budget guard (<=110%) prevent large outliers. Average adherence is 91-97% across all personas. Exact daily calorie matching would require further narrowing the calorie scoring sigma or scaling serving sizes proportionally for sub-2000 kcal profiles.

**No user feedback persistence:**

Adaptive learning operates within a single plan generation. User preferences (liked/disliked foods) are not persisted across sessions. A future version could store ratings in a local JSON and use them to bias food group weights in subsequent generations.

## TECHNICAL REFLECTION

The two most impactful design decisions were (1) the fractional gap-vector normalisation and (2) the calorie-proportional serving size. Before normalisation, absolute units caused calcium (1,000 mg RDA) to dominate fiber (25 g) by 40x in the gap-fill score, making the ANN query and ranker effectively optimise only for calcium. After fractional normalisation (gap = max(0, (RDA-actual)/RDA)), all 15 nutrients compete equally. This alone moved Priya from 0/7 to 4/7 days meeting RDA.

The five BAX-423 techniques form a sequential pipeline rather than isolated components: embeddings generate the query space, ANN recommendation retrieves candidates in that space, Bloom filter provides a safety guarantee, ranking selects the best among candidates, and streaming delivers results progressively. The adaptive gap-feedback loop ties them together into a system that improves its own recommendations as the day's plan is built.